

24BCE1567
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Probability and Statistics Lab
Assessment 6

Question 1:

Code:

```
B1=c(20, 23, 18, 17)
```

```
B2=c(19, 15, 17, 20, 16)
```

```
B3=c(21, 19, 17, 20, 16)
```

```
B4=c(15, 17, 16, 18)
```

```
group=list(B1=B1, B2=B2, B3=B3, B4=B4)
```

```
stgr=stack(group)
```

```
crd=aov(values~ind, data=stgr)
```

```
summary(crd)
```

OUTPUT:

```
> B1=c(20, 23, 18, 17)
```

```
> B2=c(19, 15, 17, 20, 16)
```

```
> B3=c(21, 19, 17, 20, 16)
```

```
> B4=c(15, 17, 16, 18)
```

```
>
```

```
> group=list(B1=B1, B2=B2, B3=B3, B4=B4)
```

```
> stgr=stack(group)
```

```
>
```

```
> crd=aov(values~ind, data=stgr)
```

```
> summary(crd)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
ind	3	21.6	7.200	1.669	0.219
Residuals	14	60.4	4.314		

QUESTION 2:

Code:

```
M1 = c(10, 10, 8, 2)
```

```
M2 = c(8, 7, 9, 8)
```

```
M3 = c(7, 12, 11, 6)
```

```
M4 = c(4, 5, 12, 11)
```

```
M5 = c(6, 11, 10, 13)
```

```

group = data.frame(M1, M2, M3, M4, M5)
temp_obs = c(as.matrix(group))
machines_f = c("M1", "M2", "M3", "M4", "M5")
bearings_f = c("B1", "B2", "B3", "B4")

Machines = rep(machines_f, each = 4)
Bearings = rep(bearings_f, times = 5)
anova_rbd = aov(temp_obs ~ Bearings + Machines)
summary(anova_rbd)

```

OUTPUT:

```

> M1 = c(10, 10, 8, 2)
> M2 = c(8, 7, 9, 8)
> M3 = c(7, 12, 11, 6)
> M4 = c(4, 5, 12, 11)
> M5 = c(6, 11, 10, 13)
>
> group = data.frame(M1, M2, M3, M4, M5)
> temp_obs = c(as.matrix(group))
> machines_f = c("M1", "M2", "M3", "M4", "M5")
> bearings_f = c("B1", "B2", "B3", "B4")
>
> Machines = rep(machines_f, each = 4)
> Bearings = rep(bearings_f, times = 5)
> anova_rbd = aov(temp_obs ~ Bearings + Machines)
> summary(anova_rbd)

```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Bearings	3	25	8.333	0.820	0.508
Machines	4	16	4.000	0.393	0.810
Residuals	12	122	10.167		

Question 3:

Code:

```

S1 = c(75, 83, 86, 73)
S2 = c(73, 72, 61, 67)
S3 = c(59, 56, 53, 62)
S4 = c(69, 70, 72, 79)
S5 = c(84, 92, 88, 95)
group = data.frame(S1, S2, S3, S4, S5)
scores = c(as.matrix(group))

```

```

students_f = c("Student1", "Student2", "Student3", "Student4", "Student5")
forms_f = c("FormA", "FormB", "FormC", "FormD")
Students = rep(students_f, each = 4)
Forms = rep(forms_f, times = 5)
anova_rbd = aov(scores ~ Forms + Students)
summary(anova_rbd)

```

OUTPUT:

```

> S1 = c(75, 83, 86, 73)
> S2 = c(73, 72, 61, 67)
> S3 = c(59, 56, 53, 62)
> S4 = c(69, 70, 72, 79)
> S5 = c(84, 92, 88, 95)
> group = data.frame(S1, S2, S3, S4, S5)
> scores = c(as.matrix(group))
> students_f = c("Student1", "Student2", "Student3", "Student4", "Student5")
> forms_f = c("FormA", "FormB", "FormC", "FormD")
> Students = rep(students_f, each = 4)
> Forms = rep(forms_f, times = 5)
> anova_rbd = aov(scores ~ Forms + Students)
> summary(anova_rbd)

```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Forms	3	43.0	14.3	0.506	0.685
Students	4	2326.7	581.7	20.572	2.65e-05 ***
Residuals	12	339.3	28.3		

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Question 4

Code:

```
B1 = c(9, 6, 9)
```

```
B2 = c(10, 13, 16)
```

```
B3 = c(7, 12, 14)
```

```

group = data.frame(B1, B2, B3)
obs_values = c(as.matrix(group))
blocks_f = c("Block1", "Block2", "Block3")
treat_f = c("T1", "T2", "T3")
Blocks = rep(blocks_f, each = 3)
Treatments = rep(treat_f, times = 3)

```

```
anova_rbd = aov(obs_values ~ Treatments + Blocks)
summary(anova_rbd)
```

OUTPUT:

```
> B1 = c(9, 6, 9)
> B2 = c(10, 13, 16)
> B3 = c(7, 12, 14)
>
> group = data.frame(B1, B2, B3)
> obs_values = c(as.matrix(group))
> blocks_f = c("Block1", "Block2", "Block3")
> treat_f = c("T1", "T2", "T3")
> Blocks = rep(blocks_f, each = 3)
> Treatments = rep(treat_f, times = 3)
>
> anova_rbd = aov(obs_values ~ Treatments + Blocks)
> summary(anova_rbd)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Treatments	2	28.67	14.333	2.688	0.182
Blocks	2	38.00	19.000	3.563	0.129
Residuals	4	21.33	5.333		